ADITYA UNIVERSITY

School of Computing

Department of Data Science

DATA VISUALIZATION

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"Global Military Fire Power Analysis" is* submitted by

G. Bhanu Teja	24B11DS072
A. Madhan	24B11DS002
M. Chethan	24B11DS128
K. Durga	25B21DS008

in partial fulfilment of the requirements for the award of the B.Tech. degree in Data Science for the academic year 2026-2027.

Project Guide
P. Srilatha, M.Tech., (Ph.D)
Assistant Professor,
Department of IT&DS

Head of the Department
Dr. M. Rajababu, Ph.D,
Associate Professor & HOD
Department of IT&DS

DECLARATION

We hereby declare that the project entitled "**Global Military Firepower Analysis**" is our original work and is submitted to **ADITYA UNIVERSITY**, **Surampalem**, in partial fulfillment of the requirements for the award of the **B.Tech. degree in Data Science**.

We further declare that this project work has not been submitted, either in full or in part, for the award of any degree or diploma in this or any other institution. We also confirm that this work has been carried out with academic honesty and complies with the **Institutional Academic Integrity and AI Usage Policy**.

Place: Surampalem

Date:

By

G. Bhanu Teja	24B11DS072
A. Madhan	24B11DS002
M. Chethan	24B11DS128
K. Durga	25B21DS008

ACKNOWLEDGEMENT

It is with immense pleasure that we express our sincere gratitude to our Cornerstone Project Guide**, P. Srilatha, Assistant Professor**, who has guided us throughout the project. Her valuable support, encouragement, and guidance have greatly contributed to the successful completion of this work.

We would like to thank our Deputy Heads of the Department, **Dr. A. Lakshmana Rao, Associate Professor** and **Mrs S. Neelima, Assistant Professor**, Department of IT & DS, for their support and valuable suggestions.

Our deepest thanks to our Head of the Department, **Dr. M. Rajababu, Associate Professor & HOD**, for his inspiration and for providing all the necessary facilities and resources required for this project.

We also extend our sincere thanks **to Dr. M.V. Rajesh, Associate Dean**, **School of Computing,** for his encouragement and support during our project.

We would like to express our sincere gratitude to **Dr. G. Suresh, Registrar, Dr. A. Ramesh, Pro Vice-Chancellor (Engineering & Sciences), Dr. S. Rama Sree, Pro Vice-Chancellor (Academics), Dr. M.B. Srinivas, Vice-Chancellor, and Dr. M. Sreenivasa Reddy, Deputy Pro-Chancellor and Management, Aditya University**, for providing excellent infrastructure and state-of-the-art laboratories.

Finally, we would like to thank the **faculty members, lab technicians, non-teaching staff, and our friends** who have directly or indirectly supported us in completing this Cornerstone project successfully.

ABSTRACT

GLOBAL MILITARY FIREPOWER ANALYSIS

The **Global Military Firepower Analysis** is an interactive data visualization project that analyses the Global Military Firepower 2025 dataset, covering 145 countries and 57 assessment factors. The dataset describes demographics and manpower, financial indicators, air, land and naval hardware, logistics, geography and natural-resource consumption. The project follows a systematic analytics workflow: data collection, cleaning and preprocessing in Microsoft Excel and Power Query, pivot-table based summarization, and dashboard development in Microsoft Power BI Desktop.

Twelve pivot tables summarize country-wise and region-wise aircraft, population and manpower, naval fleet, submarines, destroyers, rocket projectors, sea and land coverage, and coal, natural gas and oil consumption. These are presented through two interactive Power BI dashboard pages. The first page contains a country-wise aircraft bar chart, a military manpower versus total population column chart, a natural gas consumption donut chart, a coal consumption pie chart and a multi-line chart of destroyers, rocket projectors, submarines and total naval fleet. The second page contains a land-coverage map, KPI cards for military strength, foreign exchange, defence budget and water coverage, and a gauge chart for trainer aircraft, aircraft carriers and attack aircraft. Region slicers (Africa, Asia, Europe, North America, Oceania and South America) allow users to filter every visual.

The project shows how large, multi-dimensional defence data can be converted into clear and interactive visual insights that support comparison of military capability, regional strength and resource dependence across nations.

TABLE OF CONTENTS

CERTIFICATE	i
DECLARATION	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	Iv
TABLE OF CONTENTS	1
LIST OF FIGUERS	3
CHAPTER 1: INTRODUCTION	3
1.1 Introduction to the Project	3
1.2 Problem Statement	3
1.3 Objectives of the Project	3
1.4 Scope of the Project	3
CHAPTER 2: DATASET DESCRIPTION AND UNDERSTANDING	5
2.1 Dataset Source	5
2.2 Dataset Description	5
2.3 Features / Attributes Used in the Dashboard	6
2.4 Dataset Statistics / Initial Exploration	6
CHAPTER 3: DATA PREPROCESSING	7
3.1 Data Collection / Loading	7
3.2 Handling Missing Values	7
3.3 Handling Duplicate / Inconsistent Data	7
3.4 Other Project-Specific Preprocessing	7
CHAPTER 4: SYSTEM DESIGN AND METHODOLOGY	8
4.1 Proposed Methodology	8
4.2 System Workflow / Architecture	8
4.3 Flowchart	8
4.4 Tools and Technologies Used	9
4.5 Software and Hardware Requirements	10
CHAPTER 5: IMPLEMENTATION	11
5.1 Development Environment	11

5.2 Dataset Loading and Exploration	11
5.3 Preprocessing Implementation	11
5.4 Pivot Table Creation (Exploratory Data Analysis)	11
5.5 Dashboard Implementation in Power BI	17
5.6 Important Calculations	17
CHAPTER 6: DATA VISUALIZATION AND RESULTS	18
6.1 Bar Chart – Country-wise Aircraft	18
6.2 Column Chart – Military Manpower vs Total Population	18
6.3 Donut Chart – Natural Gas Consumption	19
6.4 Pie Chart – Coal Consumption	19
6.5 Line Chart – Destroyers, Rocket Projectors, Submarines and Total Naval Fleet	20
6.6 Map – Land Coverage Area of Countries	20
6.7 KPI Cards – Key Performance Indicators	21
6.8 Gauge Chart – Trainer Aircraft, Aircraft Carriers and Attack Aircraft	22
6.9 Region Slicers	23
6.10 Interpretation of Visualizations	23
CHAPTER 7: ANALYSIS OF RESULTS	24
7.1 Experimental Results	24
7.2 Result Interpretation	24
7.3 Discussion	24
CHAPTER 8: OUTPUT SCREENSHOTS	25
8.1 Input / Dataset Screenshots	25
8.2 Visualization Outputs	26
CHAPTER 9: CONCLUSION AND FUTURE ENHANCEMENTS	27
9.1 Conclusion	27
9.2 Limitations	27
9.3 Future Enhancements	27
REFERENCES	30

LIST OF FIGUERS

Fig No.	Caption / Title	Page No.
Fig 1.1	Raw dataset – Excel view (remaining columns)	Page 7
Fig 2.1	Project flowchart	Page 9
Fig 3.1	PT 1 – Country-wise Aircraft (pivot table)	Page 13
Fig 3.2	PT 2 – Region-wise Population and Manpower (pivot table)	Page 13
Fig 3.3	PT 3 – Region-wise Aircraft (pivot table)	Page 13
Fig 3.4	PT 4 – Asia – Country-wise Population and Military (pivot table)	Page 14
Fig 3.5	PT 5 – Country-wise Attack Aircraft (pivot table)	Page 14
Fig 3.6	PT 6 – Country-wise Naval Fleet (pivot table)	Page 15
Fig 3.7	PT 7 – Country-wise Military Data (pivot table)	Page 15
Fig 3.8	PT 8 – Country-wise Sea Coverage (pivot table)	Page 16
Fig 3.9	PT 9 – Country-wise Land Coverage (pivot table)	Page 16
Fig 3.10	PT 10 – Country-wise Coal Data (pivot table)	Page 17
Fig 3.11	PT 11 – Country-wise Natural Gas Consumption (pivot table)	Page 17
Fig 3.12	PT 12 – Country-wise Oil Consumption (pivot table)	Page 17
Fig 4.1	Country Wise Aircraft chart	Page 19
Fig 4.2	Military Manpower vs Total Population chart	Page 19
Fig 4.3	Natural Gas Consumption donut chart	Page 20
Fig 4.4	Coal Consumption pie chart	Page 20
Fig 4.5	Naval multi-line chart	Page 21
Fig 4.6	Land Coverage map	Page 21
Fig 4.7	Military Strength KPI card	Page 22
Fig 4.8	Foreign Exchange KPI card	Page 22
Fig 4.9	Defence Budget KPI card	Page 22
Fig 4.10	Water Coverage KPI card	Page 23
Fig 4.11	Gauge chart	Page 23
Fig 4.12	Region slicer buttons	Page 24
Fig 5.1	Input dataset	Page 26
Fig 5.2	Dashboard Page 1 – full view	Page 27
Fig 5.3	Dashboard Page 2 – full view	Page 27

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Project

Global defence capability is determined by many interacting factors: population and manpower, active and reserve forces, air, land and naval hardware, financial strength, geography and access to energy resources. Raw datasets with hundreds of rows and dozens of columns are hard to interpret in tabular form. The Global Military Power Analytical Dashboard converts the Global Military Firepower 2025 dataset into a set of interactive visualizations built with Microsoft Excel pivot tables and Microsoft Power BI.

The dashboard lets analysts, students and decision-makers compare countries and regions, identify leaders in each military domain and understand how resource consumption relates to military scale.

1.2 Problem Statement

In the modern geopolitical landscape, military power is determined not merely by raw troop counts, but by how nations combine manpower, technology, naval and air capability, finances and natural resources. Raw datasets and surface-level rankings fail to reveal per-capita militarization, domain specialization and regional concentration of power. Without an interactive visualization system, evaluating military readiness, force allocation and resource dependence across nations remains fragmented and opaque. This project addresses the need by building an interactive dashboard that makes these comparisons simple and visual.

1.3 Objectives of the Project

1. To clean and preprocess the Global Military Firepower 2025 dataset, including removal of duplicate country entries and validation of data types.
2. To create pivot tables in Excel that summarize the data by country and by region.
3. To design interactive Power BI dashboards using map, bar, column, line, pie, donut, gauge and KPI-card visuals.
4. To provide region-wise slicers so that every visual can be filtered interactively.
5. To derive insights on global military strength, regional dominance, naval and air capability and energy consumption.

1.4 Scope of the Project

The project covers 145 countries across six regions (Africa, Asia, Europe, North America, Oceania and South America). It includes data cleaning, twelve pivot-table summaries and two

dashboard pages. The analysis is descriptive and uses a single-year (2025) snapshot of the dataset; it does not include forecasting or machine-learning models.

CHAPTER 2: DATASET DESCRIPTION AND UNDERSTANDING

2.1 Dataset Source

The dataset used is the Global Military Firepower 2025 dataset (global_military_firepower_2025.xlsx), derived from the Global Firepower (GFP) 2025 ranking available at globalfirepower.com. It is provided as an Excel workbook.

2.2 Dataset Description

The dataset contains 145 rows (one per country) and 57 columns (assessment factors) covering military, economic and logistical capabilities. The columns can be grouped into the following categories.

Category	Attributes
Demographics & Manpower	Total population, total military manpower, fit for service, annual military-age population, active personnel, reserve personnel, paramilitary forces
Financial Indicators	Defence budget (USD), external debt, purchasing power parity (PPP), foreign exchange and gold reserves
Air Power	Total military aircraft, fighter, attack, transport, trainer, special-mission and tanker aircraft, total helicopters, attack helicopters
Land Power	Tanks, armored fighting vehicles, self-propelled artillery, towed artillery, rocket projectors
Naval Power	Total naval fleet, fleet tonnage, aircraft carriers, helicopter carriers, submarines, destroyers, frigates, corvettes, coastal patrol craft, mine-warfare craft
Logistics & Geography	Serviceable airports, labor force, ports and terminals, merchant marine, railway and roadway coverage, land area, coastline, shared borders
Natural Resources	Production, consumption and proven reserves of oil, natural gas and coal

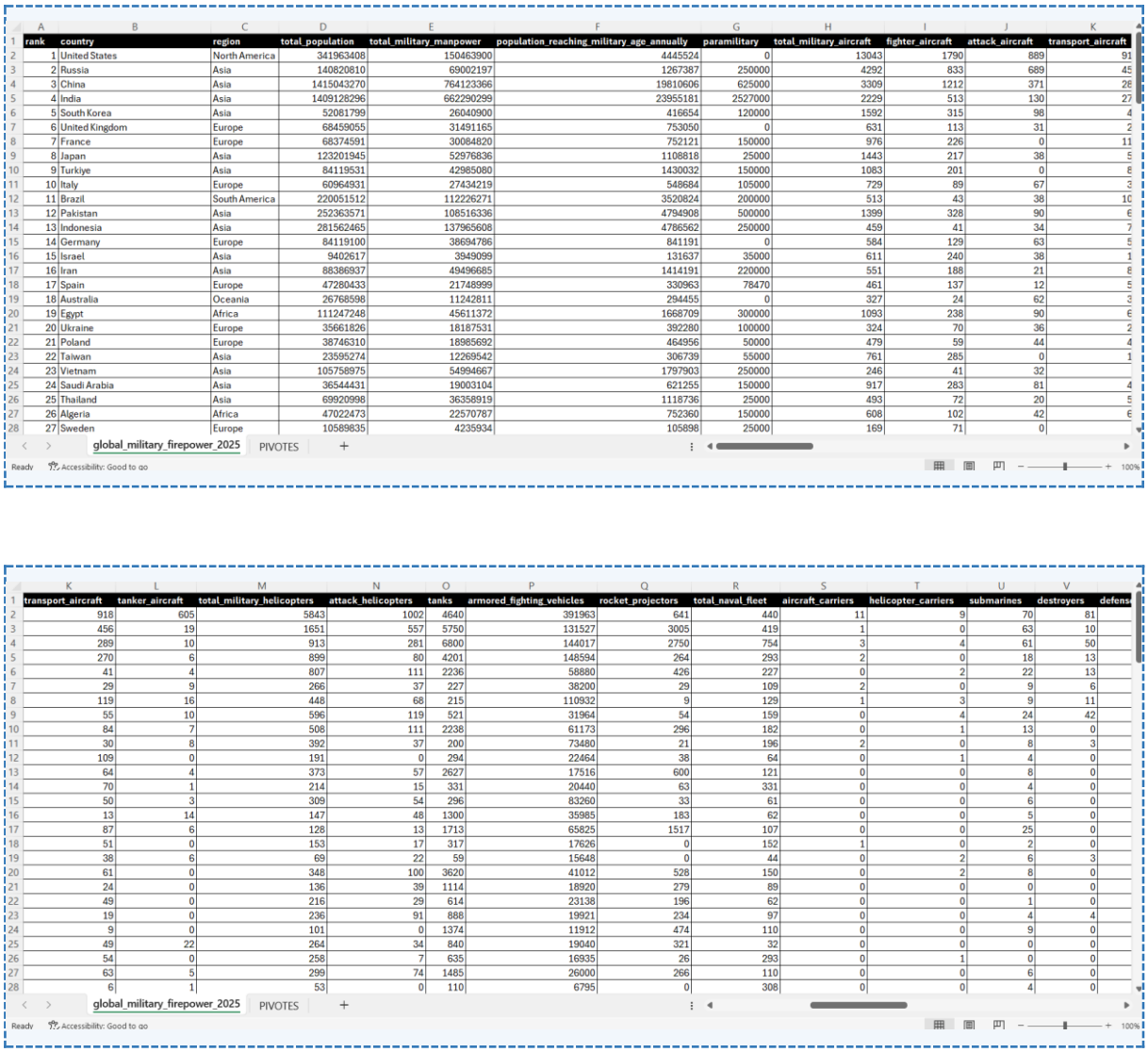
2.3 Features / Attributes Used in the Dashboard

Attribute	Used For
country, region	Grouping, map and region slicers
total_population, total_military_manpower	Manpower vs population chart; Military Strength KPI
fighter_aircraft, attack_aircraft, trainer_aircraft, aircraft_carriers	Aircraft bar chart and gauge chart
destroyers, rocket_projectors, submarines, total_naval_fleet	Multi-line naval chart
land coverage, water (sea) coverage	Land-coverage map; Water Coverage KPI

Attribute	Used For
defence budget, foreign exchange reserves	Defence Budget and Foreign Exchange KPIs
coal_consumption_mt, natural_gas_consumption_cum, oil consumption	Pie chart, donut chart and pivot tables

2.4 Dataset Statistics / Initial Exploration

The workbook contains 145 countries and 57 attributes. Initial exploration involved checking the structure of the sheet, column names and data types, identifying duplicate country records in the Asia region, and checking for blank or zero values. Countries with a legitimate zero value (for example, no submarines or no aircraft carriers) were retained because they represent real data points.



rank	country	region	total_population	total_military_manpower	population_reaching_military_age_annually	paramilitary	total_military_aircraft	fighter_aircraft	attack_aircraft	transport_aircraft
1	United States	North America	341963408	150463900	4445524	0	13043	1790	889	91
2	Russia	Asia	140820810	69002197	1267387	250000	4292	833	689	45
3	China	Asia	1415043270	764123366	19810606	625000	3309	1212	371	26
4	India	Asia	1409128296	662290299	23955181	2527000	2229	513	130	27
5	South Korea	Asia	52081799	26040900	416654	120000	1592	315	98	4
6	United Kingdom	Europe	68459055	31491165	753050	0	631	113	31	2
7	France	Europe	68374591	30084820	752121	150000	976	226	0	11
8	Japan	Asia	123201945	52976836	1108818	25000	1443	217	38	5
9	Turkiye	Asia	84119531	42985080	1430032	150000	1083	201	0	8
10	Italy	Europe	60964931	27434219	548684	105000	729	89	67	3
11	Brazil	South America	220051512	112226271	3520824	200000	513	43	38	10
12	Pakistan	Asia	252363571	108516336	4794908	500000	1399	328	90	6
13	Indonesia	Asia	281562465	137965608	4786562	250000	459	41	34	7
14	Germany	Europe	84119100	38694786	841191	0	584	129	63	5
15	Israel	Asia	9402617	3949099	131637	35000	611	240	38	1
16	Iran	Asia	88386937	49496685	1414191	220000	551	188	21	8
17	Spain	Europe	47280433	21748999	330963	78470	461	137	12	5
18	Australia	Oceania	26768598	11242811	294455	0	327	24	62	3
19	Egypt	Africa	111247248	45611372	1668709	300000	1093	238	90	6
20	Ukraine	Europe	35661826	18187531	392280	100000	324	70	36	2
21	Poland	Europe	38746310	18985692	464956	50000	479	59	44	4
22	Taiwan	Asia	23595274	12269542	306739	55000	761	285	0	1
23	Vietnam	Asia	105758975	54994667	1797903	250000	246	41	32	
24	Saudi Arabia	Asia	36544431	19003104	621255	150000	917	283	81	4
25	Thailand	Asia	69920998	36358919	1118736	25000	493	72	20	5
26	Algeria	Africa	47022473	22570787	752360	150000	608	102	42	6
27	Sweden	Europe	10589835	4235934	105898	25000	169	71	0	

transport_aircraft	tanker_aircraft	total_military_helicopters	attack_helicopters	tanks	armored_fighting_vehicles	rocket_projectors	total_naval_fleet	aircraft_carriers	helicopter_carriers	submarines	destroyers	defens
918	605	5843	1002	4640	391963	641	440	11	9	70	81	
456	19	1651	557	5750	131527	3005	419	1	0	63	10	
289	10	913	281	6800	144017	2750	754	3	4	61	50	
270	6	899	80	4201	148594	264	293	2	0	18	13	
41	4	807	111	2236	58880	426	227	0	2	22	13	
29	9	266	37	227	38200	29	109	2	0	9	6	
119	16	448	68	215	110932	9	129	1	3	9	11	
55	10	596	119	521	31964	54	159	0	4	24	42	
84	7	508	111	2238	61173	296	182	0	1	13	0	
30	8	392	37	200	73480	21	196	2	0	8	3	
109	0	191	0	294	22464	38	64	0	1	4	0	
64	4	373	57	2627	17516	600	121	0	0	8	0	
70	1	214	15	331	20440	63	331	0	0	4	0	
50	3	309	54	296	83260	33	61	0	0	6	0	
13	14	147	48	1300	35985	183	62	0	0	5	0	
87	6	128	13	1713	65825	1517	107	0	0	25	0	
51	0	153	17	317	17626	0	152	1	0	2	0	
38	6	69	22	59	15648	0	44	0	2	6	3	
61	0	348	100	3620	41012	528	150	0	2	8	0	
24	0	136	39	1114	18920	279	89	0	0	0	0	
49	0	216	29	614	23138	196	62	0	0	1	0	
19	0	236	91	888	19921	234	97	0	0	4	4	
9	0	101	0	1374	11912	474	110	0	0	9	0	
49	22	264	34	840	19040	321	32	0	0	0	0	
54	0	258	7	635	16935	26	293	0	1	0	0	
63	5	299	74	1485	26000	266	110	0	0	6	0	
6	1	53	0	110	6795	0	308	0	0	4	0	

Fig 1.1 Raw dataset – Excel view (remaining columns)

CHAPTER 3: DATA PREPROCESSING

3.1 Data Collection / Loading

The Global Military Firepower 2025 dataset was downloaded in Excel format and opened in Microsoft Excel. The file was later loaded into Power BI Desktop through Power Query for modelling and visualization.

3.2 Handling Missing Values

All columns were inspected for blank or null cells. Values that are zero because a country does not possess an asset were retained, since replacing them would distort comparisons. Numeric columns that were stored as text were converted to numeric format so that they could be aggregated correctly.

3.3 Handling Duplicate / Inconsistent Data

Duplicate country entries for the Asia region were identified and removed so that each country appears exactly once. Column headers were standardized to a consistent naming convention (lower-case names with underscores, such as total_military_manpower) to enable error-free import into Power BI.

3.4 Other Project-Specific Preprocessing

- Region column validated so that every country maps to exactly one of the six regions.
- Numeric fields formatted as whole numbers or decimals as appropriate.
- Pivot-table-ready layout prepared, with one row per country and one column per attribute.
- Data loaded into Power BI using Power Query, and DAX used for measures where required.

CHAPTER 4: SYSTEM DESIGN AND METHODOLOGY

4.1 Proposed Methodology

The methodology starts with collecting the Global Military Firepower 2025 dataset and cleaning it in Excel. Twelve pivot tables summarize the data from different angles. The cleaned data and pivot summaries are then loaded into Power BI, where the dashboard visuals, KPI cards and region slicers are created and arranged on a custom military-themed background.

4.2 System Workflow / Architecture

Step	Stage	Description
1	Global Military Firepower 2025 dataset	Source Excel workbook (145 rows × 57 columns)
2	Data collection & loading	Open in Excel / load into Power BI
3	Data cleaning & preprocessing	Remove duplicates, validate types, standardize columns
4	Pivot table creation	Twelve country-wise and region-wise summaries
5	Data modelling in Power BI	Relationships, measures and formatting
6	Dashboard development	Page 1 (charts) and Page 2 (map, KPIs, gauge)
7	Interactive analysis	Region slicers, cross-filtering, tooltips

4.3 Flowchart

Insert the project flowchart below (Dataset → Cleaning → Pivot Tables → Power BI → Dashboard → Insights).



Fig 2.1 Project flowchart

4.4 Tools and Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning, removal of duplicates, pivot tables
Power Query	Data loading and transformation inside Power BI
DAX	Measures and calculations in Power BI
Microsoft Power BI Desktop	Interactive dashboard creation
CSV / XLSX	Data source formats
Google Chrome / Microsoft Edge	Dataset download and report publishing

4.5 Software and Hardware Requirements

Software Requirements	
Operating System	Windows 10 / 11
Data Visualization Tool	Microsoft Power BI Desktop
Data Source	CSV / Excel dataset
Data Cleaning	Power Query, DAX and MS Excel
Web Browser	Google Chrome or Microsoft Edge

Hardware Requirements	
Processor	Intel Core i3 or higher (i5 recommended)
RAM	Minimum 8 GB (16 GB recommended)
Storage	20 GB free disk space
Display	1366 × 768 resolution or higher
Internet	Required for downloading datasets and publishing reports
Input devices	Standard keyboard and mouse

CHAPTER 5: IMPLEMENTATION

5.1 Development Environment

The project was developed on Windows using Microsoft Excel for data cleaning and pivot tables and Microsoft Power BI Desktop for the dashboard. Power Query handled data transformation, and DAX was used for calculations where needed.

5.2 Dataset Loading and Exploration

The Excel workbook was opened and the structure of all 57 columns reviewed. The sheet was then converted into a clean tabular layout suitable for pivot tables.

5.3 Preprocessing Implementation

Duplicate Asia country rows were removed using Excel's Remove Duplicates and manual verification, column headers were standardized and data types were validated, as described in Chapter 3.

5.4 Pivot Table Creation (Exploratory Data Analysis)

Twelve pivot tables were created to explore the data from different analytical angles. Each pivot table is documented below with its purpose, followed by space for the screenshot.

PT #	Pivot Table	Purpose
PT 1	Country-wise Aircraft	Compares air-force size across nations and feeds the Country Wise Aircraft bar chart.
PT 2	Region-wise Population and Manpower	Shows regional demographic strength and feeds the Military Manpower vs Total Population chart.
PT 3	Region-wise Aircraft	Shows air power by continent.
PT 4	Asia – Country-wise Population and Military	Deep dive into the military demographics of Asia.
PT 5	Country-wise Attack Aircraft	Shows offensive air capability.
PT 6	Country-wise Naval Fleet	Compares naval strength across nations.
PT 7	Country-wise Military Data	Multi-metric comparison; feeds the multi-line naval chart.
PT 8	Country-wise Sea Coverage	Shows maritime territory; feeds the Water Coverage KPI.
PT 9	Country-wise Land Coverage	Shows territorial land area; feeds the land-coverage map.
PT 10	Country-wise Coal Data	Energy-resource analysis; feeds the coal consumption pie chart.
PT 11	Country-wise Natural Gas Consumption	Energy-resource analysis; feeds the natural gas donut chart.

PT #	Pivot Table	Purpose
PT 12	Country-wise Oil Consumption	Energy-resource analysis for oil dependence.

PT 1: Country-wise Aircraft

Configuration: Rows: Country. Values: aircraft counts (such as fighter aircraft).

Purpose: Compares air-force size across nations and feeds the Country Wise Aircraft bar chart.

Row Labels	Sum of fighter_aircraft
United States	1790
China	1212
Russia	833
India	513
North Korea	368
Pakistan	328
South Korea	315
Taiwan	285
Saudi Arabia	283
Israel	240
Egypt	238
France	226
Japan	217
Turkiye	201
Iran	188
Greece	178
Spain	137
Germany	129
United Kingdom	113
Syria	104
Algeria	102
Qatar	102
Singapore	100
United Arab Emirates	99
Italy	89
Morocco	83
Thailand	72

Fig 3.1 PT 1 – Country-wise Aircraft (pivot table)

PT 2: Region-wise Population and Manpower

Configuration: Rows: Region. Values: total population and total military manpower.

Purpose: Shows regional demographic strength and feeds the Military Manpower vs Total Population chart.

Row Labels	Sum of total_population	Sum of total_military_manpower
Africa	1399584580	574955853
Asia	4859172457	2398493043
Europe	600684356	279401800
North America	579256127	259790963
Oceania	31929809	13462132
South America	441366055	216527800
Grand Total	**7911993384**	**3742631591**

Fig 3.2 PT 2 – Region-wise Population and Manpower (pivot table)

PT 3: Region-wise Aircraft

Configuration: Rows: Region. Values: aircraft counts.

Purpose: Shows air power by continent.

Row Labels	Sum of total_military_aircraft
Africa	4099
Asia	24545
Europe	6796
North America	14123
Oceania	375
South America	2201
Grand Total	**52139**

Fig 3.3 PT 3 – Region-wise Aircraft (pivot table)

PT 4: Asia – Country-wise Population and Military

Configuration: Filter: Region = Asia. Rows: Country. Values: population and military manpower.

Purpose: Deep dive into the military demographics of Asia.

Row Labels	Sum of total_population	Sum of total_military_manpower
Asia	**4859172457**	**2398493043**
Afghanistan	40121552	15647405
Armenia	2976765	1666988
Azerbaijan	10650239	5218617
Bahrain	1566888	846120
Bangladesh	168697184	82661620
Bhutan	884546	203446
Cambodia	17063669	8190561
China	1415043270	764123366
Georgia	4900961	2744538
India	1409128296	662290299
Indonesia	281562465	137965608
Iran	88386937	49496685
Iraq	42083436	17675043
Israel	9402617	3949099
Japan	123201945	52976836
Jordan	11174024	3463947
Kazakhstan	20260006	9117003
Kuwait	3138355	1726095
Kyrgyzstan	6172101	3147772
Laos	7953556	3499565
Lebanon	5364482	2092148
Malaysia	34564810	16245461
Mongolia	3281676	1870555
Myanmar	57527139	30489384
Nepal	31122387	15249970
North Korea	26298666	9467520
Oman	3901992	1482757
Pakistan	252363571	108516336
Philippines	118277063	50859137
Qatar	2552088	638022
Russia	140820810	69002197
Saudi Arabia	36544431	19003104
Singapore	6028459	3918498
South Korea	52081799	26040900
Sri Lanka	21982608	10551652

Fig 3.4 PT 4 – Asia – Country-wise Population and Military (pivot table)

PT 5: Country-wise Attack Aircraft

Configuration: Rows: Country. Values: attack aircraft.

Purpose: Shows offensive air capability.

Row Labels	Sum of attack_aircraft
United States	889
Russia	689
China	371
India	130
North Korea	114
South Korea	98
Pakistan	90
Egypt	90
Saudi Arabia	81
Italy	67
Germany	63
Australia	62
Belarus	50
Poland	44
Algeria	42
Brazil	38
Japan	38
Israel	38
Iraq	36
Ukraine	36
Indonesia	34
Mexico	33
Norway	32
Vietnam	32
United Kingdom	31
Sudan	27
Turkmenistan	27
Myanmar	26
Philippines	25
Kazakhstan	25
Colombia	24

Fig 3.5 PT 5 – Country-wise Attack Aircraft (pivot table)

PT 6: Country-wise Naval Fleet

Configuration: Rows: Country. Values: total naval fleet.

Purpose: Compares naval strength across nations.

Row Labels	Sum of total_naval_fleet
China	754
United States	440
Russia	419
Indonesia	331
Sweden	308
Thailand	293
India	293
Sri Lanka	270
Finland	264
Colombia	233
Myanmar	232
South Korea	227
North Korea	207
Italy	196

Fig 3.6 PT 6 – Country-wise Naval Fleet (pivot table)

PT 7: Country-wise Military Data

Configuration: Rows: Country. Values: submarines, rocket projectors, naval fleet and destroyers.

Purpose: Multi-metric comparison; feeds the multi-line naval chart.

Row Labels	Sum of rocket_projectors	Sum of total_naval_fleet	Sum of submarines	Sum of destroyers
Afghanistan	0	0	0	0
Albania	135	19	0	0
Algeria	266	110	6	0
Angola	113	31	0	0
Argentina	26	44	2	3
Armenia	100	0	0	0
Australia	0	44	6	3
Austria	0	0	0	0
Azerbaijan	268	38	4	0
Bahrain	17	64	0	0
Bangladesh	110	118	2	0
Belarus	206	0	0	0
Belgium	0	9	0	0
Beliz	0	8	0	0
Benin	0	2	0	0
Bhutan	0	0	0	0
Bolivia	0	67	0	0
Bosnia and Herzegovina	79	0	0	0
Botswana	20	0	0	0
Brazil	38	64	4	0
Bulgaria	24	36	0	0
Burkina Faso	9	0	0	0
Cambodia	463	20	0	0

Fig 3.7 PT 7 – Country-wise Military Data (pivot table)

PT 8: Country-wise Sea Coverage

Configuration: Rows: Country. Values: water / sea coverage.

Purpose: Shows maritime territory; feeds the Water Coverage KPI.

Row Labels	Sum of coastline_coverage_km	Sum of waterway_coverage_km
Afghanistan	0	1200
Albania	362	41
Algeria	998	0
Angola	1600	1300
Argentina	4989	11000
Armenia	0	0
Australia	25760	2000
Austria	0	358
Azerbaijan	713	0
Bahrain	161	0
Bangladesh	580	8370
Belarus	0	2500
Belgium	67	2043
Beliz	386	825
Benin	121	150
Bhutan	0	0
Bolivia	0	10000
Bosnia and Herzegovina	20	990
Botswana	0	0
Brazil	7491	50000
Bulgaria	354	470
Burkina Faso	0	0
Cambodia	443	3700
Cameroon	400	0
Canada	202080	636
Central African Republic	0	2800

Fig 3.8 PT 8 – Country-wise Sea Coverage (pivot table)

PT 9: Country-wise Land Coverage

Configuration: Rows: Country. Values: land area.

Purpose: Shows territorial land area; feeds the land-coverage map.

Row Labels	Sum of border_coverage_km	Sum of total_land_area_sq_km
Afghanistan	5987	652230
Albania	691	28748
Algeria	6734	2381740
Angola	5369	1246700
Argentina	11968	2780400
Armenia	1570	29743
Australia	0	7741220
Austria	2524	83871
Azerbaijan	2468	86600
Bahrain	0	760
Bangladesh	4413	148460
Belarus	3599	207600
Belgium	1297	30528
Beliz	542	22966
Benin	2123	112622
Bhutan	1136	38394
Bolivia	7252	1098581
Bosnia and Herzegovina	1543	51197
Botswana	4347	581730
Brazil	16145	8515770
Bulgaria	1806	110880
Burkina Faso	3611	274200
Cambodia	2530	181035
Cameroon	5018	475440
Canada	8893	9984670

Fig 3.9 PT 9 – Country-wise Land Coverage (pivot table)

PT 10: Country-wise Coal Data

Configuration: Rows: Country / Region. Values: coal consumption (metric tons).

Purpose: Energy-resource analysis; feeds the coal consumption pie chart.

Row Labels	Sum of coal_production_cum	Sum of coal_consumption_mt	Sum of proven_coal_reserves_cum
Africa	268132000	206273000	21237998000
Asia	7411980060	7966978000	5.51103E+11
Europe	453996000	592845000	1.32907E+11
North America	599326000	516118500	2.56734E+11
Oceania	468901000	110065000	1.57802E+11
South America	76558000	53538000	15453998000
Grand Total	**9278893060**	**9445817500**	**1.13524E+12**

Fig 5.10 PT 10 – Country-wise Coal Data (pivot table)

PT 11: Country-wise Natural Gas Consumption

Configuration: Rows: Country / Region. Values: natural gas consumption (cubic metres).

Purpose: Energy-resource analysis; feeds the natural gas donut chart.

Row Labels	Sum of natural_gas_production_cum	Sum of natural_gas_consumption_cum	Sum of proven_natural_gas_reserves_cum
Africa	2.40893E+11	1.60995E+11	1.75205E+13
Asia	2.05196E+12	2.01499E+12	1.52534E+14
Europe	2.28043E+11	4.72E+11	3.46476E+12
North America	1.24877E+12	1.13868E+12	1.57202E+13
Oceania	1.57553E+11	54007000000	3.25915E+12
South America	1.22483E+11	1.24588E+11	7.23426E+12
Grand Total	**4.04971E+12**	**3.96526E+12**	**1.99733E+14**

Fig 3.11 PT 11 – Country-wise Natural Gas Consumption (pivot table)

PT 12: Country-wise Oil Consumption

Configuration: Rows: Country. Values: oil consumption.

Purpose: Energy-resource analysis for oil dependence.

Row Labels	Sum of oil_production_bbl	Sum of oil_consumption_bbl	Sum of proven_oil_reserves_bbl
Africa	7301825	4346000	1.2418E+11
Asia	51473420	48984700	9.83519E+11
Europe	3577232	13307600	13761040000
North America	28713023	25162000	2.14515E+11
Oceania	398000	1277000	2486993000
South America	7272400	5471000	3.30652E+11
Grand Total	**98735900**	**98548300**	**1.66911E+12**

Fig 3.12 PT 12 – Country-wise Oil Consumption (pivot table)

5.5 Dashboard Implementation in Power BI

After the pivot summaries were prepared, the cleaned data was loaded into Power BI Desktop. A custom military-themed background (world map, fighter jets, battle tank and warships) was designed and each visual was placed inside a bordered panel. Six region buttons (Africa, Asia, Europe, North America, Oceania and South America) were added at the top and work as slicers. The dashboard has two pages with a consistent title bar and the Aditya University logo.

5.6 Important Calculations

- Total military manpower and total population are aggregated by country using Sum.
- Aircraft, naval and resource-consumption values are aggregated with Sum and displayed by country or region.
- KPI cards display the top two countries (by Sum of the selected metric) for Military Strength, Foreign Exchange, Defence Budget and Water Coverage.
- The gauge chart shows the combined total of trainer aircraft, aircraft carriers and attack aircraft, with the minimum and maximum axis values.

CHAPTER 6: DATA VISUALIZATION AND RESULTS

This chapter presents each chart and KPI of the Power BI dashboard individually. Paste the screenshot of each visual into the space provided under its description.

6.1 Bar Chart – Country-wise Aircraft

Chart type: Clustered column / bar chart. Axis: Country. Value: Sum of fighter aircraft.

The chart ranks countries by fighter aircraft. The United States leads with a count of about 1,800, followed by China at roughly 1,200, Russia at about 830 and India at about 500. After the top four, counts fall steeply and most nations have only a few hundred fighter aircraft or fewer, showing that air power is highly concentrated in a few countries.

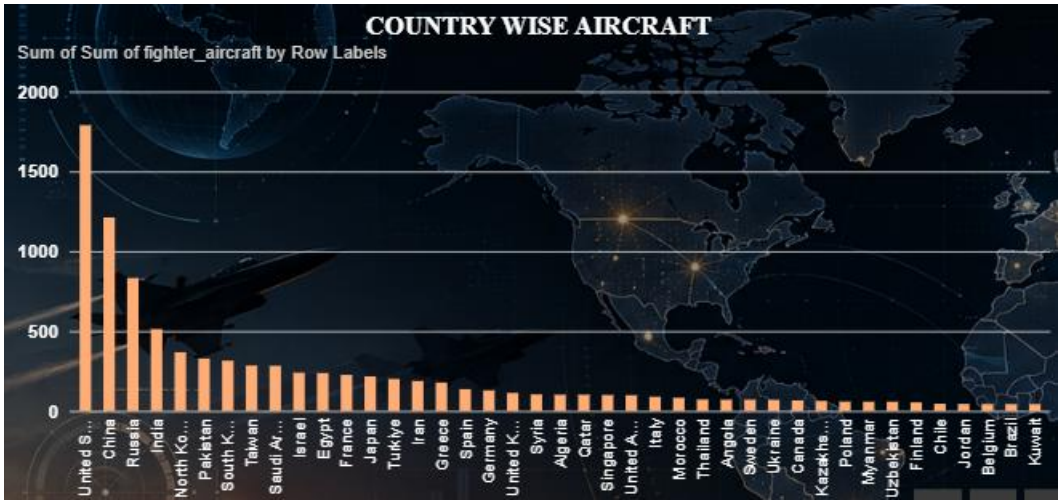


Fig 4.1 Country Wise Aircraft chart

6.2 Column Chart – Military Manpower vs Total Population

Chart type: Clustered column chart. Axis: Country. Values: Sum of total military manpower and Sum of total population.

China and India dominate both measures, with values approaching 1.5 billion on the axis for population, followed by Indonesia, Pakistan, Bangladesh and Russia. For every country, military manpower is only a fraction of total population, which highlights the difference between available manpower and total population.

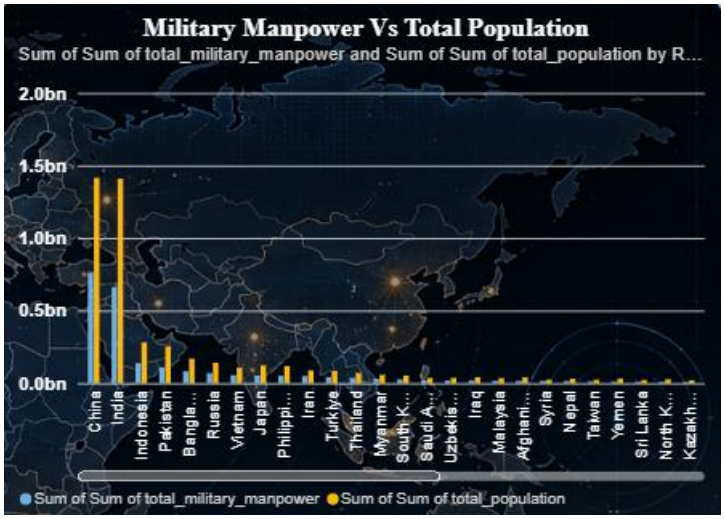


Fig 4.2 Military Manpower vs Total Population chart

6.3 Donut Chart – Natural Gas Consumption

Chart type: Donut chart. Legend: Region. Value: Sum of natural gas consumption.

Asia accounts for the largest share of natural gas consumption at about 153T (73.42%). The next regions contribute about 18T (8.43%), 16T (7.5%) and smaller shares such as 2T (0.97%) and 1T. The centre of the donut shows the total.

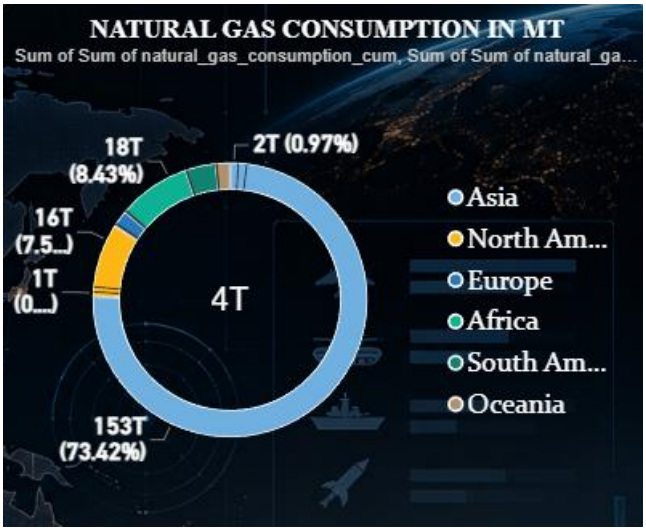


Fig 4.3 Natural Gas Consumption donut chart

6.4 Pie Chart – Coal Consumption

Chart type: Pie chart. Legend: Region. Value: Sum of coal consumption in metric tons.

Asia consumes about 8bn metric tons (84.34%) of coal, far ahead of Europe (about 6.28%) and North America (about 5.46%). The remaining regions together make up a very small share, showing the heavy dependence of Asian economies on coal.

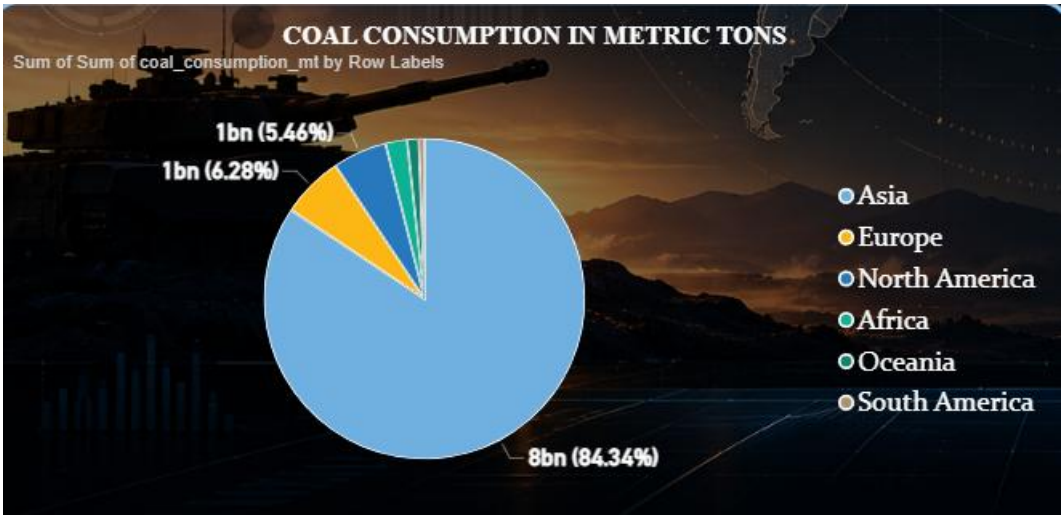


Fig 4.4 Coal Consumption pie chart

6.5 Line Chart – Destroyers, Rocket Projectors, Submarines and Total Naval Fleet

Chart type: Multi-line chart. Axis: Country. Series: Sum of destroyers, rocket projectors, submarines and total naval fleet.

Two clear peaks appear for rocket projectors (green line) for a small number of countries, while destroyers and submarines stay close to zero for most nations. Naval and rocket-artillery strength is concentrated in a few countries, and the majority of countries have very small values.

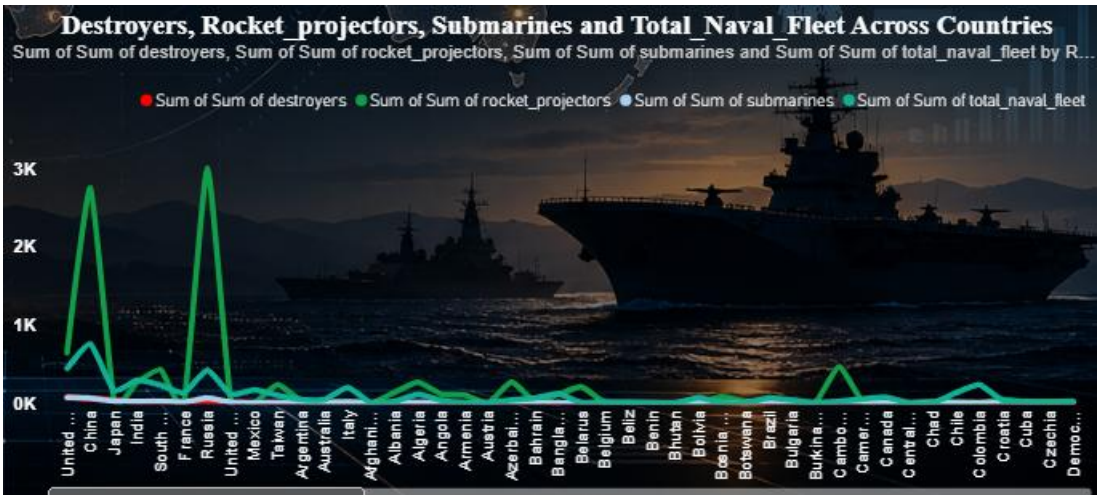


Fig 4.5 Naval multi-line chart

6.6 Map – Land Coverage Area of Countries

Chart type: Filled (ArcGIS) map. Location: Country. Colour: land coverage.

Russia is shown in the darkest shade as the country with the largest land area. China, India, Kazakhstan, Chad, DR Congo and others are shaded in lighter colours according to land area. The map gives an immediate spatial view of territorial size.



Fig 4.6 Land Coverage map

6.7 KPI Cards – Key Performance Indicators

KPI cards summarize the leading countries for four key measures. Each card is shown below with the values displayed on the dashboard.

KPI Card	First Country	Value	Second Country	Value
Military Strength (Sum of total military manpower)	China	764M	India	662M
Foreign Exchange (reserves)	China	3T	Japan	1T
Defence Budget	China	267bn	United … (name truncated in card)	895bn
Water Coverage	Brazil	50K	Russia	102K

Military Strength KPI



Fig 4.7 Military Strength KPI card

Foreign Exchange KPI



Fig 4.8 Foreign Exchange KPI card

Defence Budget KPI



Fig 4.9 Defence Budget KPI card

Water Coverage KPI



Fig 4.10 Water Coverage KPI card

6.8 Gauge Chart – Trainer Aircraft, Aircraft Carriers and Attack Aircraft

The gauge chart shows the combined sum of trainer aircraft, aircraft carriers and attack aircraft. The gauge ranges from 23 (minimum) to 22K (maximum), with the current value of about 11K at its centre and a marker at 3,758. It gives a quick view of how the combined strike and training capability compares with the maximum value.

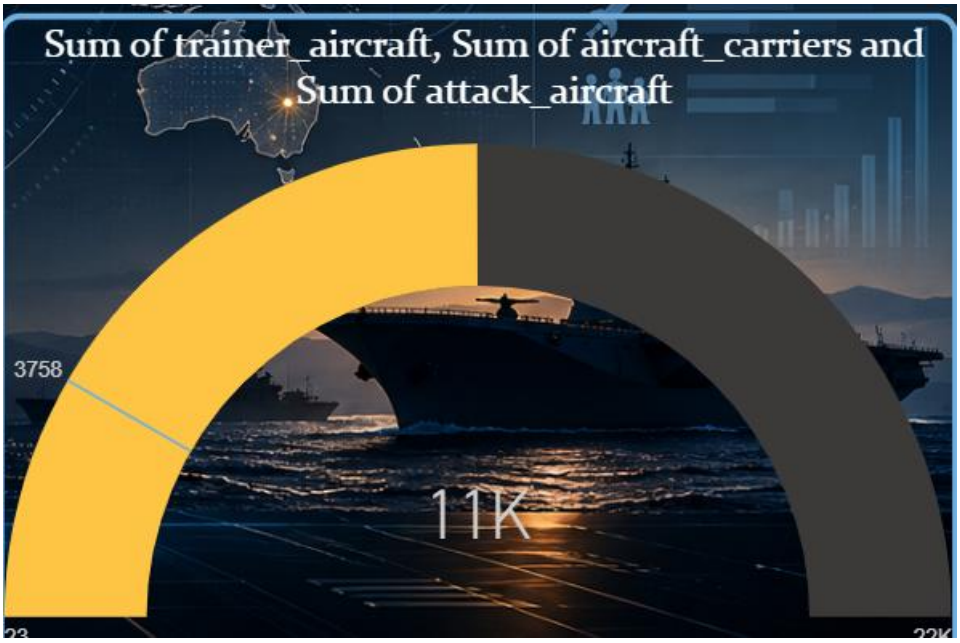


Fig 4.11 Gauge chart

6.9 Region Slicers

The six buttons at the top of both pages (Africa, Asia, Europe, North America, Oceania, South America) act as slicers. Selecting a region filters every chart, map and KPI card on the page, enabling region-level analysis.



Fig 4.12 Region slicer buttons

6.10 Interpretation of Visualizations

The bar chart compares air power, the column chart compares manpower with population, the donut and pie charts show the share of each region in natural gas and coal consumption, the line chart compares naval and rocket-artillery assets, the map shows land area, the KPI cards show the leading countries on key measures and the gauge shows combined aircraft totals. Together, they give a complete picture of global military capability and resource dependence.

CHAPTER 7: ANALYSIS OF RESULTS

7.1 Experimental Results

- **Air Power:** Air power is concentrated: the United States, China, Russia and India lead in fighter aircraft by a wide margin.
- **Manpower:** China and India have the largest population and military manpower, with China at about 764M and India at about 662M on the Military Strength KPI.
- **Energy:** Asia accounts for about 84% of coal consumption and about 73% of natural gas consumption in the dataset.
- **Naval and Rocket Forces:** Destroyers, submarines and rocket projectors are held in significant numbers by only a few countries.
- **Geography:** Russia has the largest land area; Russia (about 102K) and Brazil (about 50K) lead the Water Coverage KPI.
- **Finance:** China shows the highest foreign exchange value (about 3T) followed by Japan (about 1T). Defence budget KPI values are about 267bn for China and 895bn for the second country shown.

7.2 Result Interpretation

The results show that military capability and energy consumption are strongly concentrated in a small group of countries, mainly in Asia and North America. Larger militaries are supported by larger populations, larger industrial bases and greater energy use. A small number of countries dominate air power, naval strength and budgets, while most countries have limited capability in these domains.

7.3 Discussion

The dashboard makes it easy to compare countries and regions quickly. Region slicers support focused analysis, and the combination of charts, a map and KPI cards allows users to move from a global overview to specific countries. The project shows how business-intelligence tools can turn complex defence statistics into understandable visual insights.

CHAPTER 8: OUTPUT SCREENSHOTS

8.1 Input / Dataset Screenshots

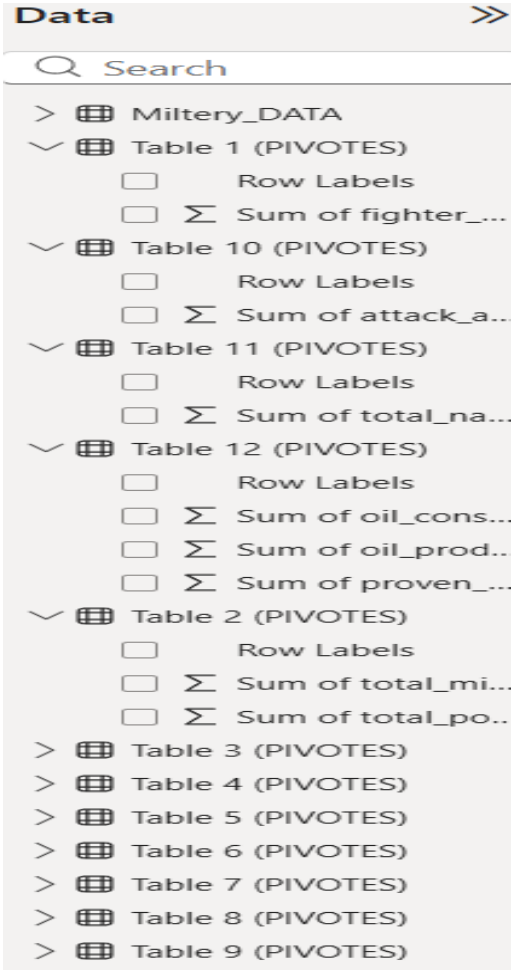


Fig 5.1 Input dataset

8.2 Visualization Outputs

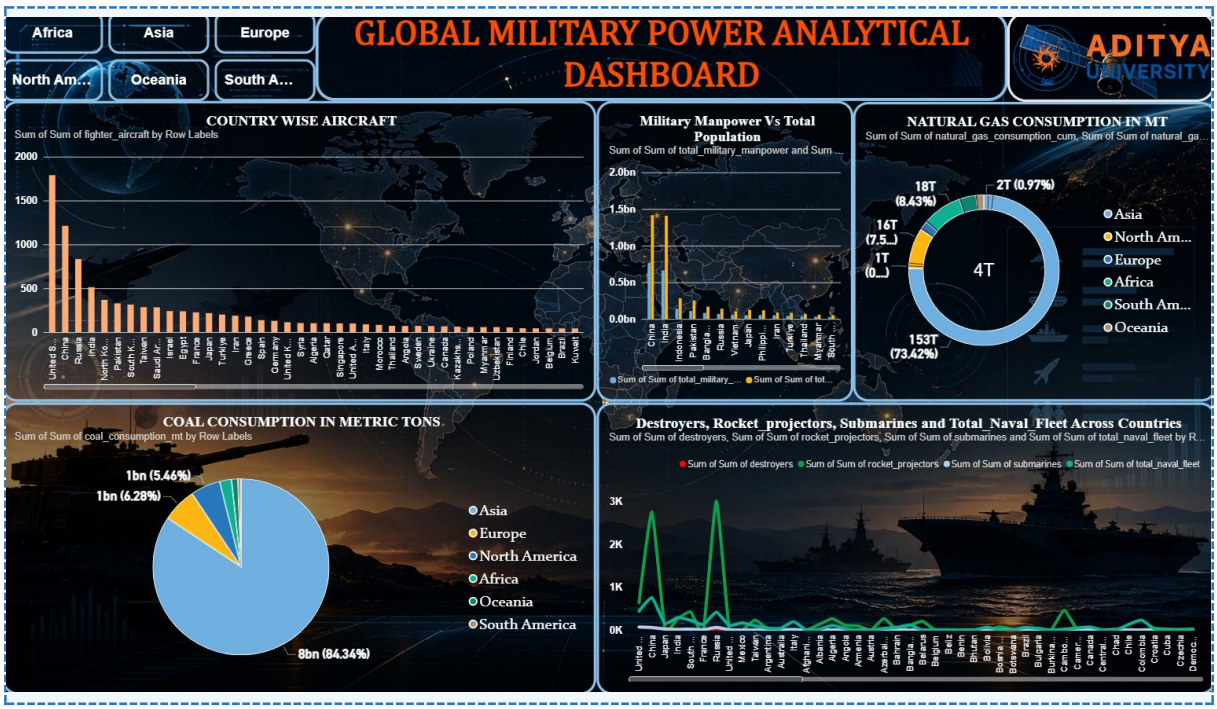


Fig 5.2 Dashboard Page 1 – full view

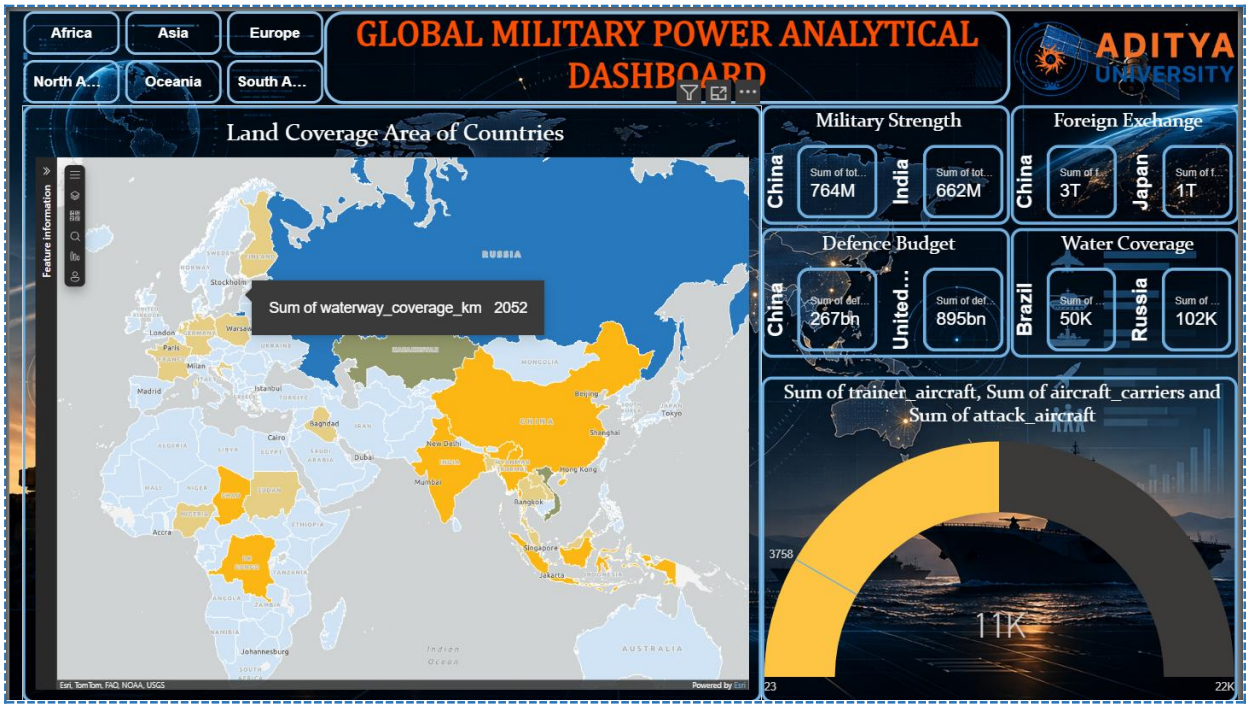


Fig 5.3 Dashboard Page 2 – full view

CHAPTER 9: CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Global Military Power Analytical Dashboard successfully analyzes the Global Military Firepower 2025 dataset using Microsoft Excel and Power BI. Data cleaning produced a reliable dataset, twelve pivot tables summarized the data from several perspectives, and two dashboard pages present country and region level insights through bar, column, line, pie, donut and gauge charts, a land-coverage map and KPI cards, all filterable by region. The project demonstrates a complete analytics workflow from raw data to interactive visual storytelling.

9.2 Limitations

- Uses a single-year (2025) snapshot and does not show trends over time.
- Descriptive analysis only; no predictions.
- Depends on the quality and completeness of the Global Firepower data.
- The proposed What-If alliance builder and per-capita readiness metrics are not part of the final dashboard.
- Some chart labels are truncated for countries with long names because of limited space.

9.3 Future Enhancements

- Add historical Global Firepower data (for example 2020–2025) to show trends over time.
- Add a What-If parameter for building hypothetical alliances and comparing blocs.
- Add per-capita militarization and readiness (active vs reserve) indices using DAX.
- Include defence budget efficiency and scatter-plot analysis.
- Publish the report to Power BI Service for sharing and collaboration.
- Introduce machine-learning models for forecasting rankings.

REFERENCES

1. Global Firepower – 2025 Military Strength Ranking. https://www.globalfirepower.com
2. Microsoft – Power BI Documentation. https://learn.microsoft.com/en-us/power-bi/
3. Microsoft – Excel PivotTable Support. https://support.microsoft.com/en-us/excel
4. Tufte, E. R. (1983). The Visual Display of Quantitative Information. Graphics Press.
5. Few, S. (2012). Show Me the Numbers: Designing Tables and Graphs to Enlighten. Analytics Press.
6. Kirk, A. (2019). Data Visualisation: A Handbook for Data Driven Design. SAGE Publications.